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## Unit “ 3 : Session “ 4 : SLO “ 2 | Stack Applications “ Infix to Postfix Conversion

### ■ I. Algorithm: Infix to Postfix Using a Stack

- Step 1: Scan the infix expression from left to right.
- Step 2: If the scanned symbol is an OPERAND, add it to the postfix expression.
- Step 3: If the scanned symbol is '(', PUSH it onto the stack.
- Step 4: If the scanned symbol is ')', POP and output all symbols until '(' is found; discard '('.
- Step 5: If the scanned symbol is an OPERATOR:
  - a) If the stack is empty or top is '(', push the operator.
  - b) Else, pop all operators from the stack whose precedence is GREATER THAN OR EQUAL to the scanned operator, output them, then push the scanned operator.
  - c) Note: '^' is right-associative “ pop only operators of strictly higher precedence.
- Step 6: Repeat Steps 1-5 until the entire infix expression is scanned.
- Step 7: Pop and output all remaining operators from the stack.
- Step 8: EXIT

### ■ II. Example: $A + B * C \rightarrow ABC*+$

- Scan A -> output A. Stack: empty
- Scan + -> push. Stack: +
- Scan B -> output B. Output: AB
- Scan \* -> \* has higher precedence than + -> push. Stack: + \*
- Scan C -> output C. Output: ABC
- End -> pop \* and +. Output: ABC\*+